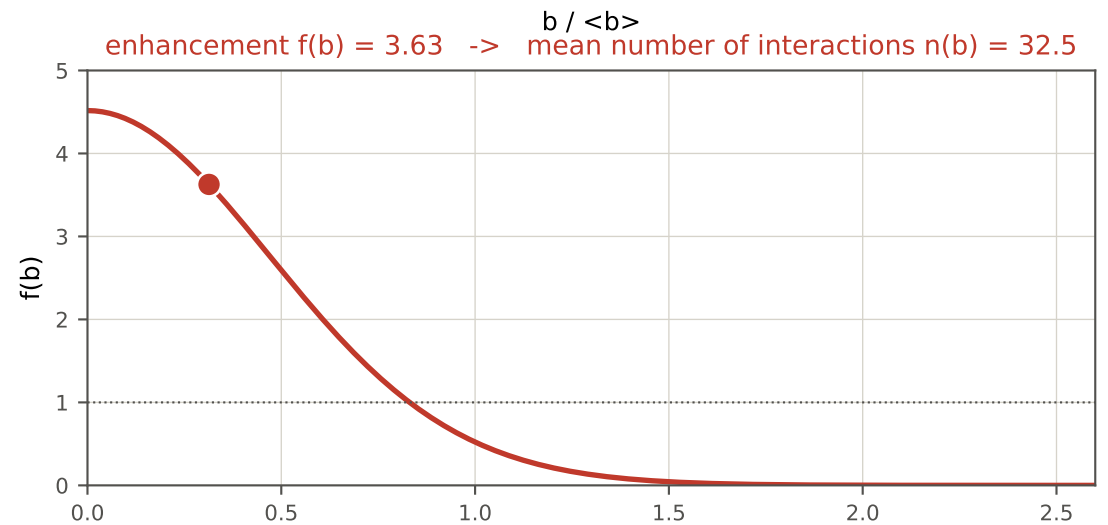
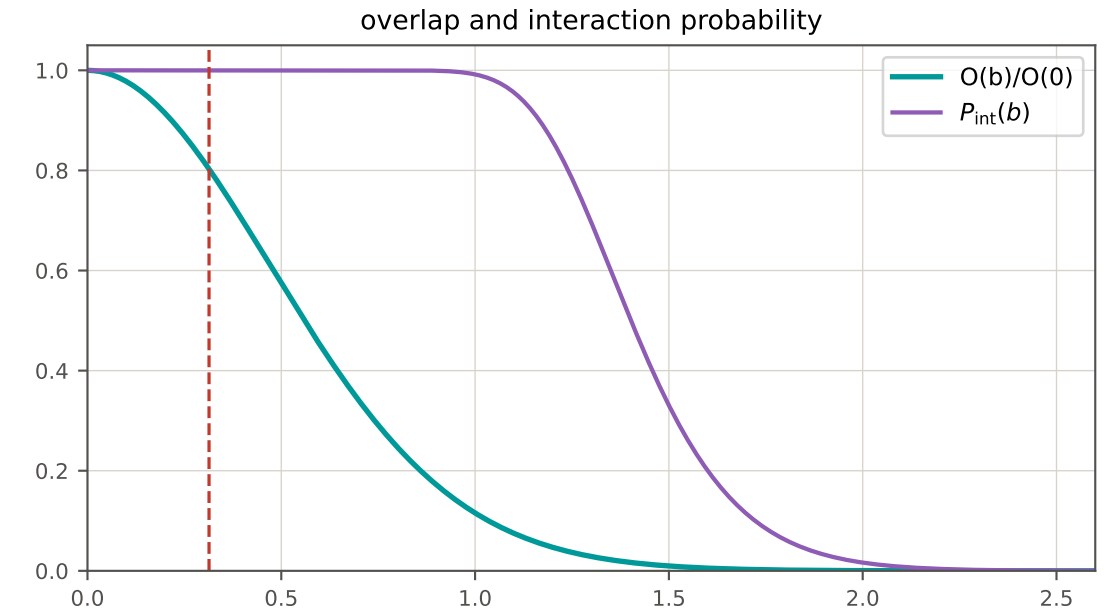
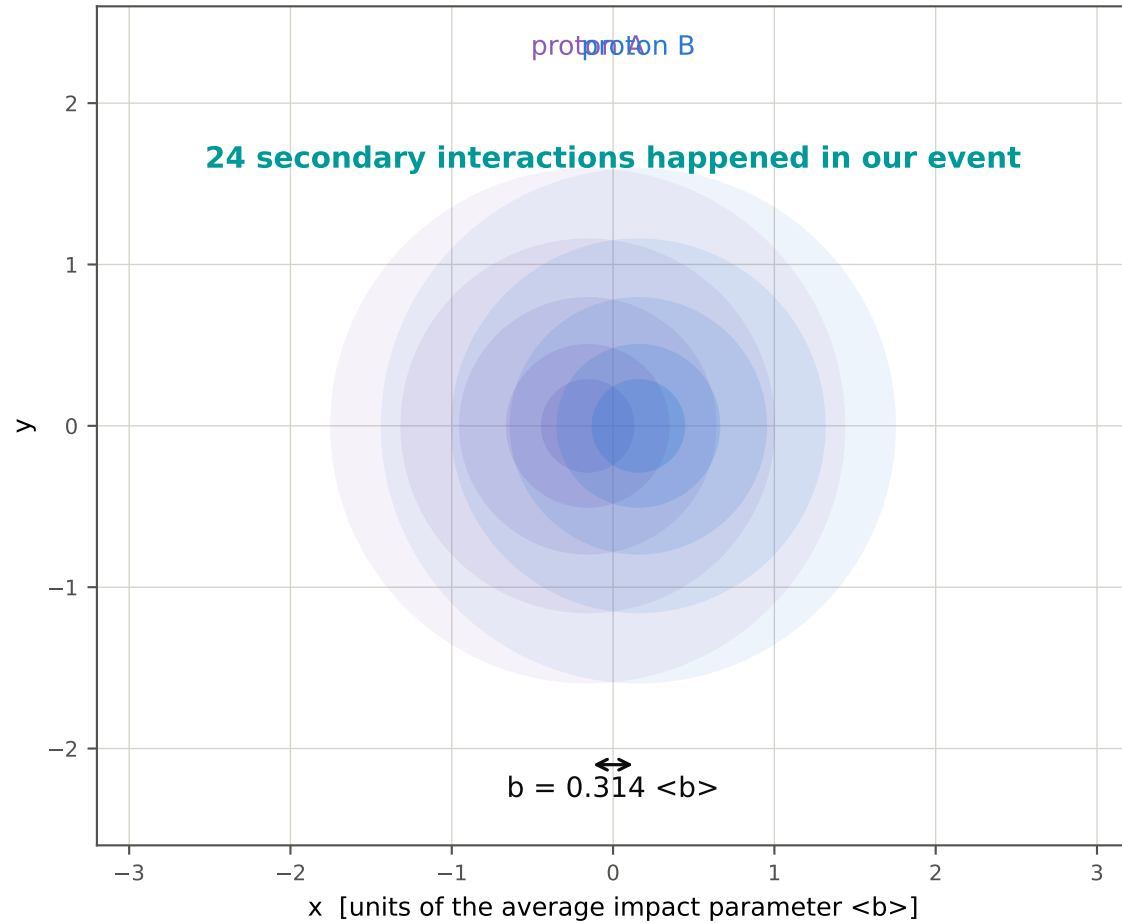


Scene 1: two protons meet at impact parameter b

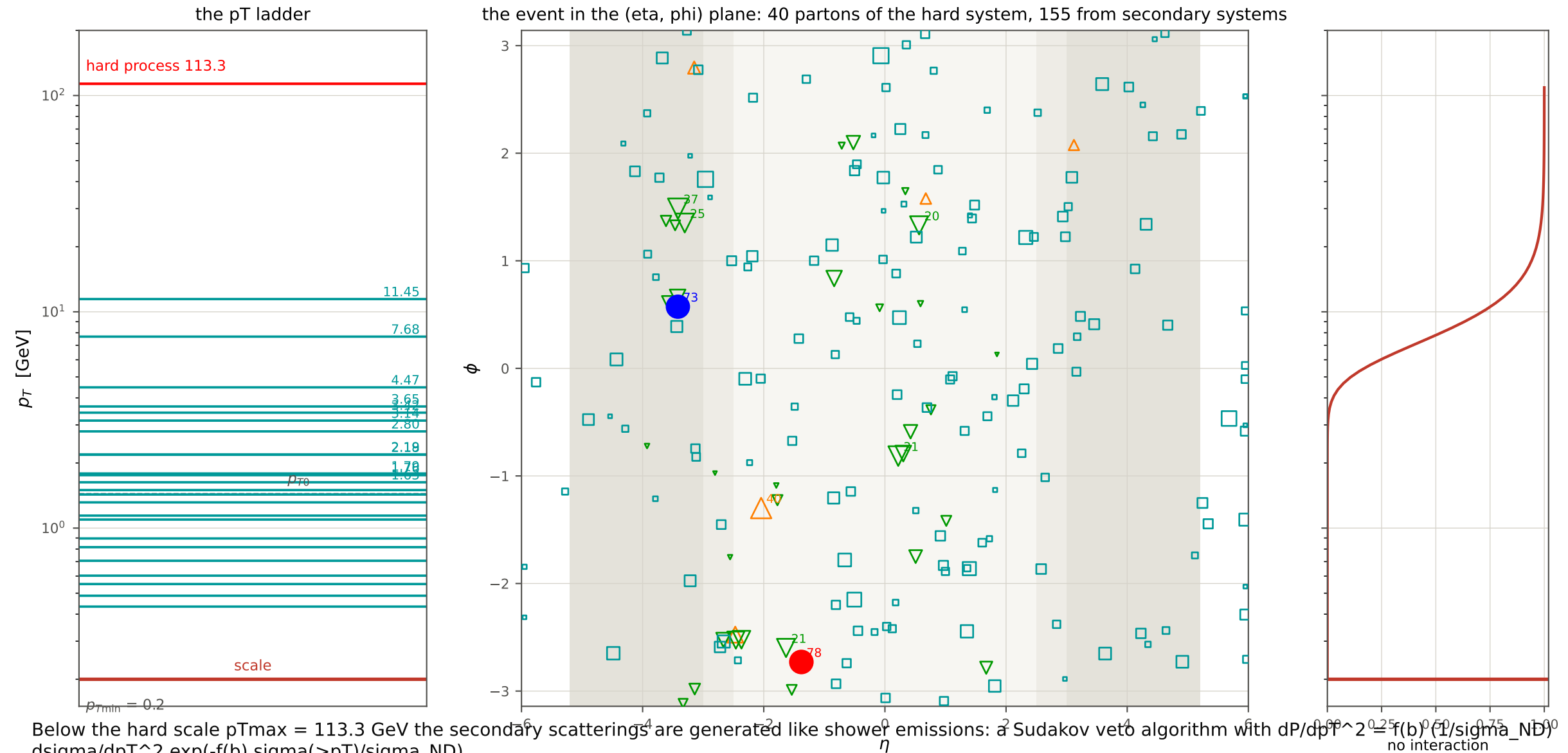
standalone_multiparton_interactions.py



The matter of each proton is a double Gaussian (CP5: 63 % of it in a core of 0.76 of the radius). The chance of a parton-parton $b / \langle b \rangle$ interaction is the overlap $O(b)$ of the two profiles; the mean number of interactions is $n(b) = f(b) \langle n \rangle$ with $\langle n \rangle = \sigma_{\text{int}}/\sigma_{\text{ND}} = 8.97$. An event with a hard process is more central than average: b is drawn with weight $O(b)$. Our event sits at $b = 0.3136 \langle b \rangle$, $f = 3.63$ (the CMSSW event: 0.3136, 3.709).

Scene 2: the chain of interactions runs down in pT

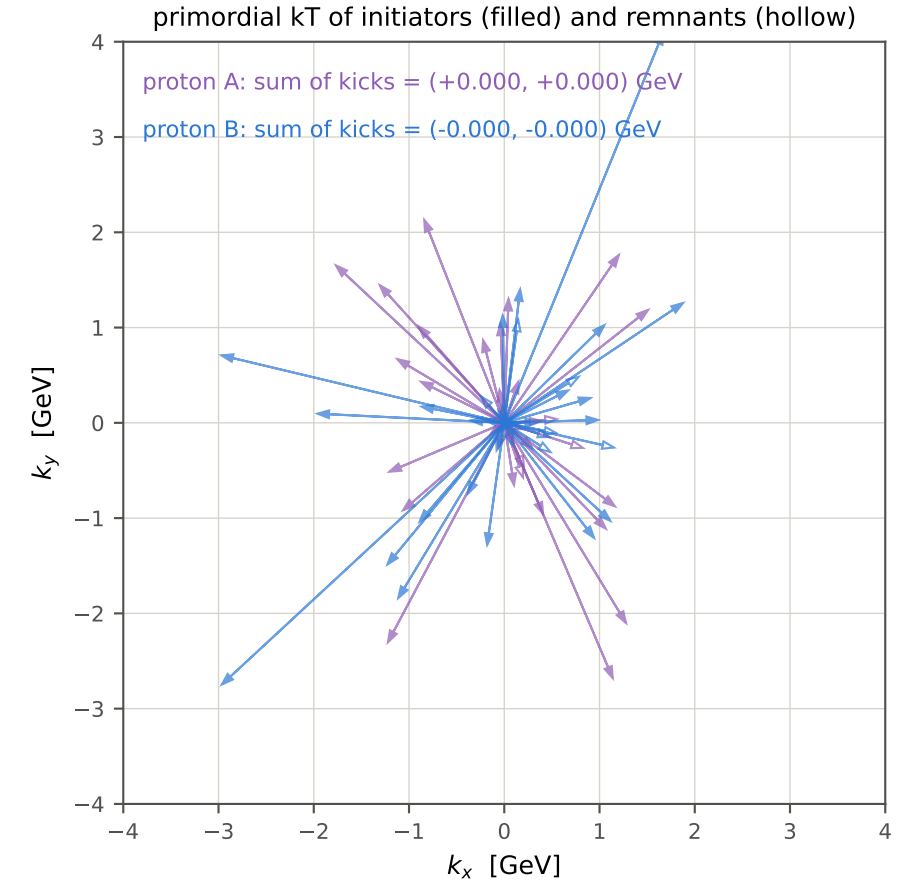
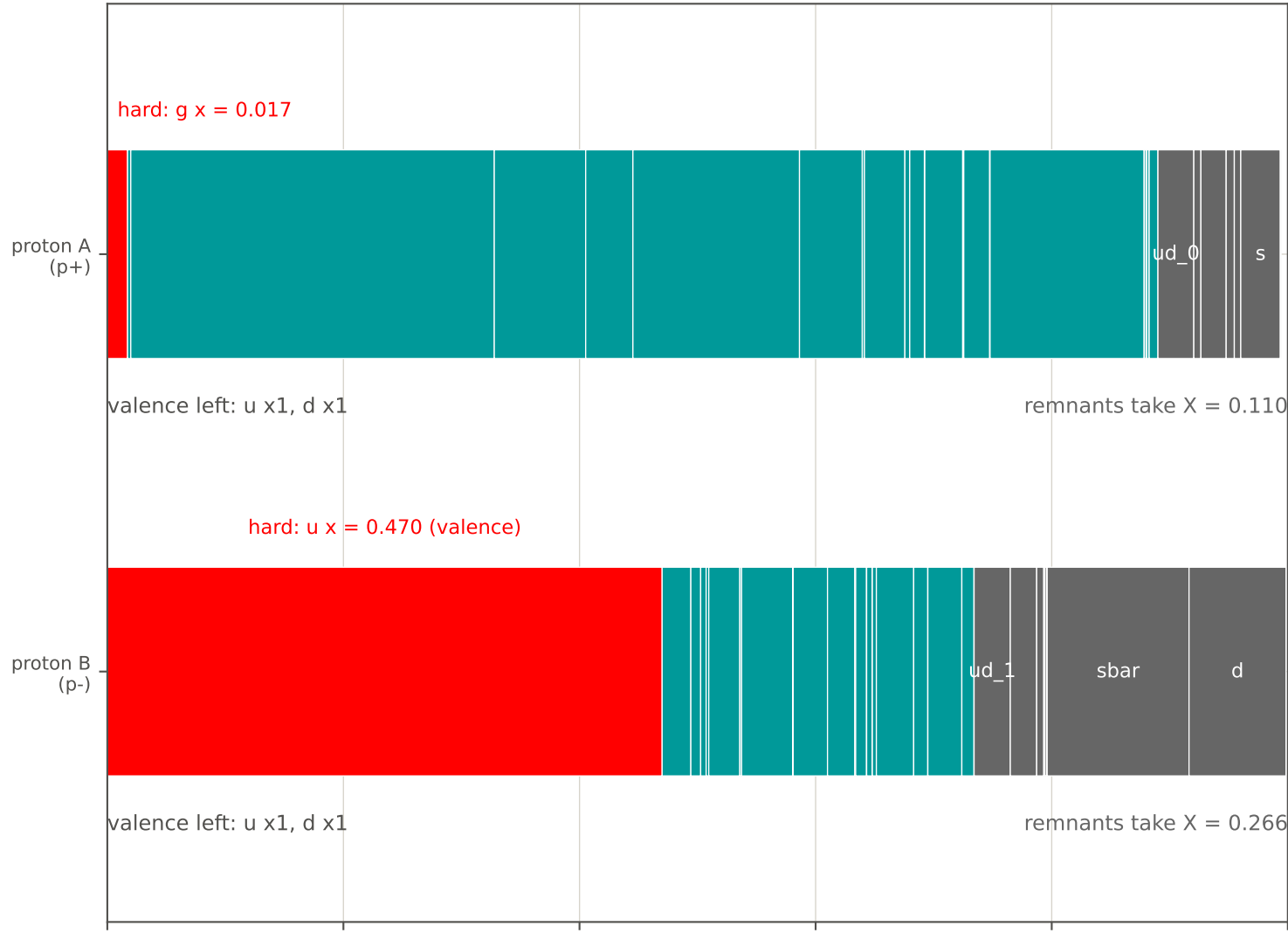
standalone_multiparton_interactions.py



Scene 3: what is left of the protons: the beam remnants

standalone_multiparton_interactions.py

interaction 25 of 25: the protons are being used up



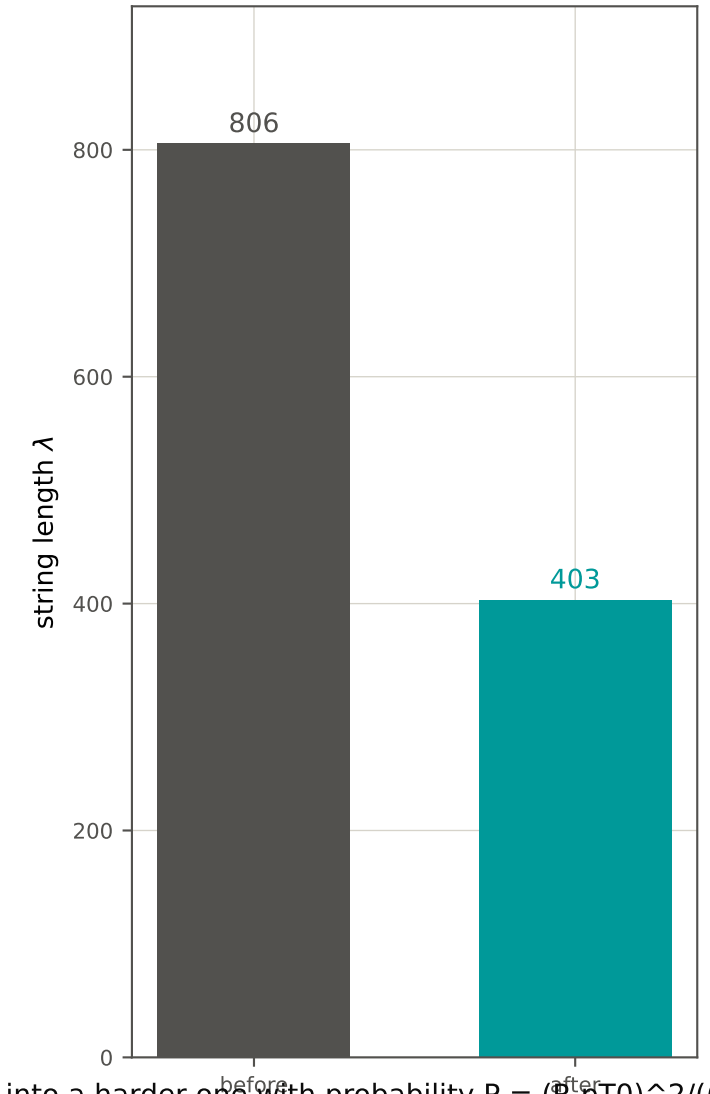
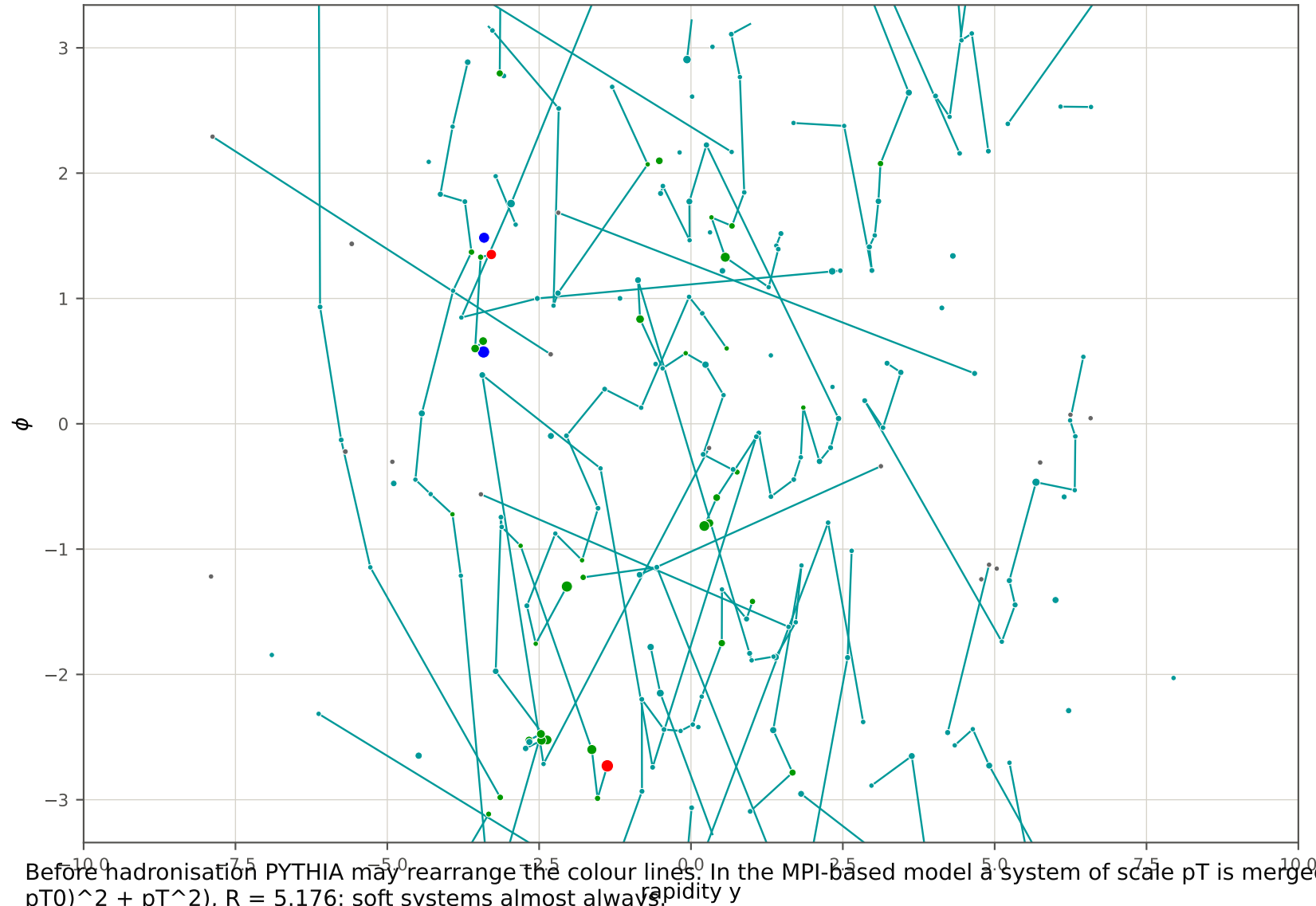
Every interaction takes a momentum fraction x out of a proton; the PDFs are rescaled to what is left, a kicked-out valence quark is subtracted from the valence content and a kicked-out sea quark leaves a companion. Proton A gave away 25 partons ($X = 0.110$ left), proton B 25 ($X = 0.266$).

The rest becomes the remnant partons (A: ud_0, u, u, ubar, sbar, s, ubar; B: ud_1, u, u, sbar, u, sbar, ubar, sbar, d); every parton gets a primordial kT and the kicks cancel within each proton.

Scene 4: colour reconnection shortens the strings

standalone_multiparton_interactions.py

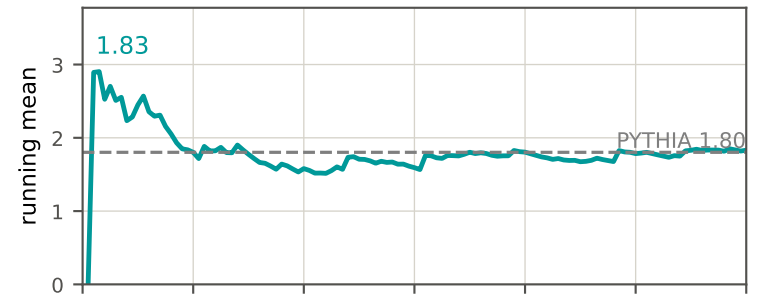
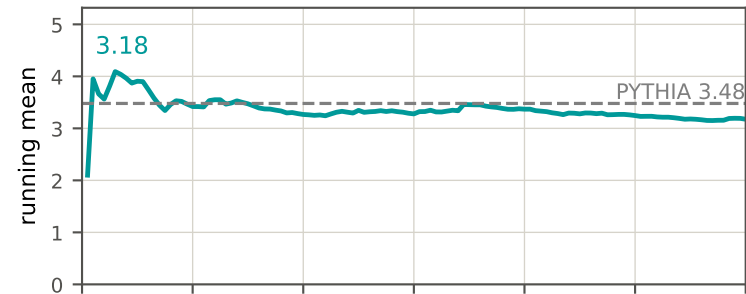
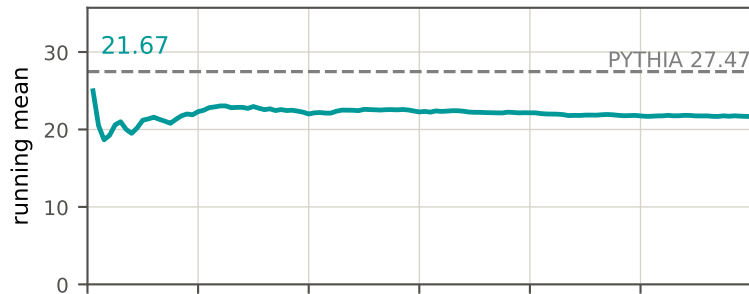
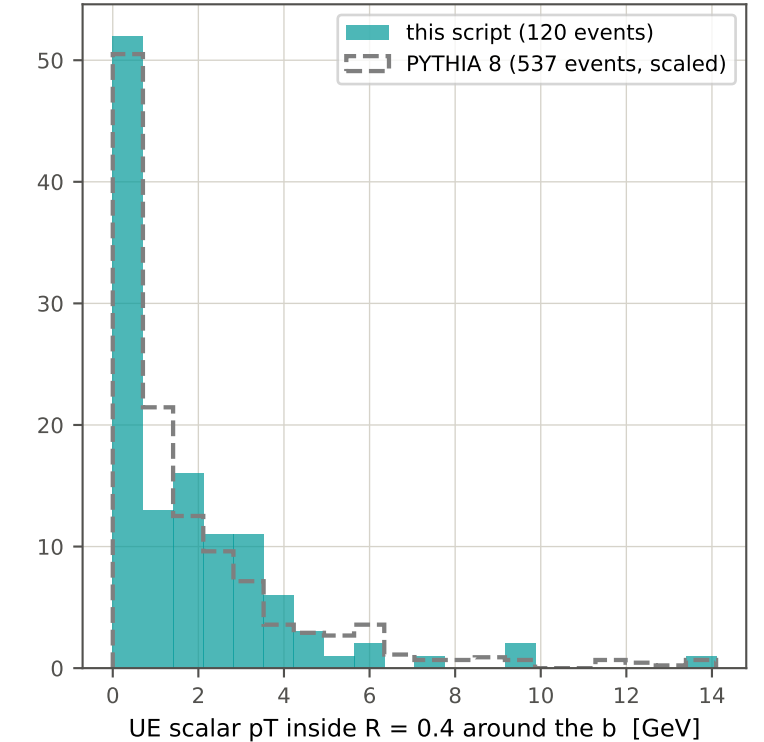
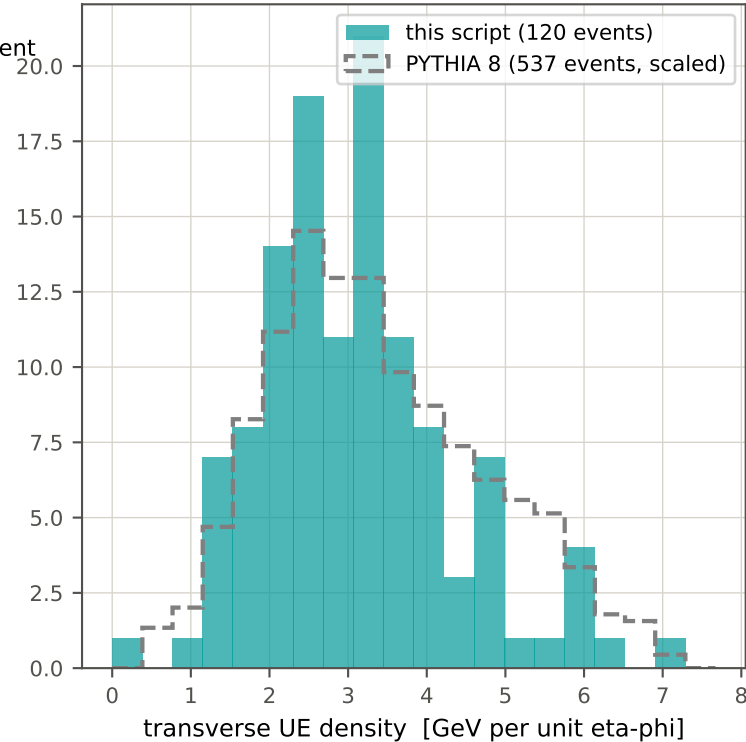
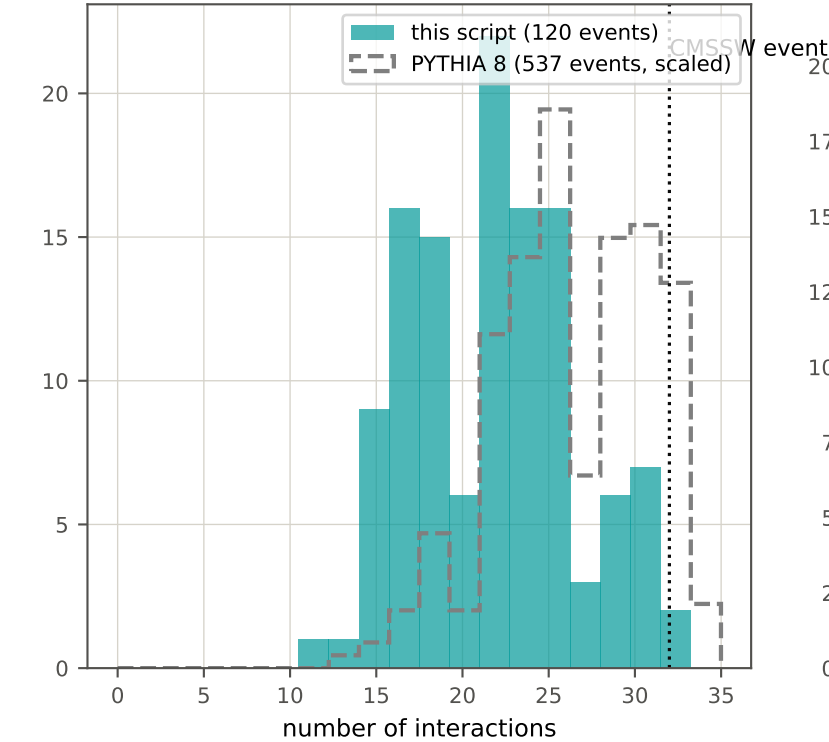
colour strings (lines) between the final-state partons: before (fading out) and after reconnection (fading in)



Before hadronisation PYTHIA may rearrange the colour lines. In the MPI-based model a system of scale p_T is merged into a harder one with probability $P = (R p_{T0})^2 / ((R p_{T0})^2 + p_T^2)$, $R = 5.176$: soft systems almost always. Its gluons are inserted into the dipoles of the harder system where they disturb it least, so the total string length $\lambda = \sum \ln(1 + m_{ij}^2/m_0^2)$ drops: 806 $\rightarrow$ 403 here, 23 of 24 systems merged, 49 $\rightarrow$ 51 strings.

Scene 5: the same event dressed again and again

standalone_multiparton_interactions.py



Seeds 1.. at $b = 0.3136$: the number of interactions, the transverse underlying-event density (partons not from the hard system, $|\eta| < 2.5$, 60-120 degrees from the leading b) and the extra scalar pT inside $R = 0.4$ around the b quark fluctuate strongly from event to event; their means settle towards the values PYTHIA 8 gives for the same hard event (standalone_mpi_stats_fixedb.stats.json).